

Alejandro Zamorano Reyes

Ph.D. in Physics

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RESEARCH INTERESTS

Self-organized criticality and avalanche dynamics; dynamics on complex and evolving networks; non-equilibrium statistical mechanics; stochastic simulation; nonlinear dynamics and pattern formation; solar flares, coronal heating and magnetic reconnection; magnetohydrodynamics and space plasmas; multifractal and time-series analysis; information-theoretic constraints on predictability.

EDUCATION

Doctor of Philosophy in Physics 2022 – 2026

Universidad de Chile, Faculty of Sciences, Santiago, Chile.

Thesis: *Analysis of Complex Networks in Systems with Self-Organized Criticality Applied to Solar Flares*

Advisors: Dr. Víctor Muñoz and Dr. Denisse Pastén.

Defended August 2026. Funded by ANID National Ph.D. Scholarship #21231335.

Bachelor of Science in Physics (Licenciatura en Ciencias, mención Física) 2021

Universidad de Chile, Faculty of Sciences, Santiago, Chile.

GRADUATE COURSEWORK

Universidad de Chile doctoral programme.

Core sequence

- Statistical Mechanics
- Electrodynamics
- Quantum Mechanics II
- Classical Mechanics
- English for Graduate Studies; English Proficiency Examination

Elective courses

- Introduction to Complex Networks
- Introduction to Complex Dynamical Systems
- Workshop on Complex Dynamical Systems II
- Topics in Nonlinear Physics
- Nonlinearity, Disorder and Non-Hermiticity in Discrete Systems (course and colloquium)
- Introduction to Fluid Dynamics
- Introduction to Plasma Physics
- Space Plasmas: Past, Present and Future
- Research Seminar I
- Oral Communication, Science and Interdisciplinarity

PUBLICATIONS

Peer-reviewed journal articles

1. **Zamorano, A.** & Muñoz, V. (2026). Effect of rewiring for a sandpile model on a directed network. *Chaos, Solitons & Fractals*, **210**, 118629.
2. **Zamorano, A.**, Morales, L., Pastén, D. & Muñoz, V. (2025). Lu and Hamilton model for solar flares over a rewiring complex network. *The Astrophysical Journal*, **994**(2), 257. doi:10.3847/1538-4357/ae1a70. Open access.

3. Flández, E., **Zamorano, A.** & Muñoz, V. (2025). A 22-yr cycle of the network topology of solar active regions. *The Astrophysical Journal*, **980**(1), 4.

Manuscripts in preparation

4. **Zamorano, A.**, Morales, L. & Muñoz, V. Temporal statistics of solar flares in a rewired Lu–Hamilton model. To be submitted to *The Astrophysical Journal*.
5. **Zamorano, A.**, Pastén, D., Morales, L. & Muñoz, V. Analysis of complex networks on unstable nodes in cellular automata models for solar flares.
6. **Zamorano, A.**, Pastén, D., Lepreti, F. & Muñoz, V. Dynamic rewiring in running sandpiles: a multifractal analogy of the Kuramoto model.
7. Villaroel-Sepúlveda, N., **Zamorano, A.** Beyond Plummer: Self-consistent superstatistical Vlasov–Poisson equilibria with continuous mass spectra.

RESEARCH EXPERIENCE

Research Assistant — Department of Physics, Universidad de Chile 2018 – 2026

FONDECYT #1201967 (2024 – present), with Dr. Víctor Muñoz

Time Series and Complex Network Approaches for Nonlinear Processes in Space Plasmas

FONDECYT #1161711 (2018 – 2024), with Dr. Víctor Muñoz

Nonlinear Phenomena in Magnetized Plasmas: A Study on Solitons and Geomagnetic Storms

- Developed sandpile and cellular-automaton models on directed complex networks whose links evolve, with degree conservation and guaranteed dissipation paths, and derived the resulting avalanche statistics.
- Identified a topological transition detected by the Gini coefficient of the load distribution and by betweenness centrality, but invisible to the avalanche size distribution and to the mean geodesic distance; characterised the asymptotic directed acyclic structure draining into sinks.
- Extended the Parker and Lu–Hamilton frameworks for coronal heating so that magnetic reconnection is represented explicitly as a change of connectivity between flux tubes, and quantified the resulting transition between scale-free and exponential energy-release statistics.
- Studied how the local connectivity of the configuration preceding an event constrains both the spatial pattern of energy deposition and the magnitude of the event.
- Characterised the evolution of solar active-region magnetic topology across a full 22-year magnetic cycle from SDO/HMI magnetograms, with flare statistics from the HEK catalogue and RHESSI.
- Implemented all models from scratch in C++ (serial), with ensembles of 10^7 events, finite-size scaling, and multifractal and waiting-time analysis in Python.

FONDECYT #1210353 (2022), with Dr. Marcel Clerc

Transition to Complex Spatiotemporal Dynamics: Theory & Experiments

- Numerical and analytical work on pattern formation and the route to spatiotemporal complexity in continuous dissipative systems.

RESEARCH VISITS

Università della Calabria, Rende, Italy

2025

Hosts: Prof. Vincenzo Carbone and Prof. Fabio Lepreti. Multifractal characterisation of intermittent energy dissipation in driven threshold models. Funded by an ANID internship grant.

Universidad de Buenos Aires, Argentina

2025

Host: Dr. Laura Morales. Avalanche models of solar flares. Funded by a PSPC–ACAL short internship grant.

ADDITIONAL TRAINING

Non-Equilibrium Statistical Physics

2026

Prof. Christopher Jarzynski (University of Maryland). Intensive course hosted by the Faculty of Chemical and Pharmaceutical Sciences, Universidad de Chile.

Introduction to Fusion Energy and Plasma Physics

2024

Princeton University, Princeton Plasma Physics Laboratory (online).

CONFERENCE PRESENTATIONS

Invited talks

- *From Sandpiles to Complex Networks to Solar Flares*. Università della Calabria, Rende, Italy (2025).
- *A Solar Pasta: from Magnetic Field Lines to Solar Flares*. Física en las Palmeras Seminar, Chile (2024 and 2025).

Contributed talks

- Conference on Complex Systems — Physics of Self-Organization in Complex Systems (2025).
- XXIV Simposio Chileno de Física, SOCHIFI (2024).
- 21st International Congress on Plasma Physics, ICPP (2024).
- Primer Congreso Chileno de Física de Plasmas (2023).
- SOCHIFI Symposium (2022).

Poster presentations

- 9th Asia-Pacific Conference on Plasma Physics, AAPPS-DPP (2025).
- 5th Meeting of the Italian Solar and Heliospheric Community, SoHe-5 (2025).
- Conference on Complex Systems (2025).
- 17th Latin American Workshop on Plasma Physics (2025).
- Dynamics Days Latin America and the Caribbean (2024).
- XIV Latin American Conference on Space Geophysics, COLAGE (2024).
- International Workshop on Statistical Physics, IWOSP (2023).
- Primer Congreso Chileno de Física de Plasmas (2023).
- European Physical Society Conference (2022).
- 11th International Conference on Complex Networks and their Applications (2022).
- XIII Latin American Conference on Space Geophysics, COLAGE (2022).
- 6th Asia-Pacific Conference on Plasma Physics, AAPPS-DPP (2022).
- International Workshop on Statistical Physics, IWOSP (2021).
- 10th International Conference on Complex Networks and their Applications (2021).
- Latin American Conference on Space Geophysics, COLAGE (2021).
- Sociedad Chilena de Física, SOCHIFI (2020).

TEACHING EXPERIENCE

Teaching Assistant — Department of Physics, Universidad de Chile

2019 – 2024

Graduate courses

- Electrodynamics
- Statistical Mechanics
- Classical Mechanics

Undergraduate courses

- Classical Mechanics
- Quantum Mechanics I
- Electrodynamics
- Statistical Mechanics
- Thermodynamics
- Mathematical Methods II
- Programming and Advanced Numerical Methods
- Contemporary Physics II
- Introduction to Mechanics

GRANTS, FELLOWSHIPS AND AWARDS

- **ANID National Ph.D. Scholarship** #21231335, Chile 2023 – 2026
- **Internship Grant** (ANID), Università della Calabria, Italy 2025
- **Short Internship Grant** (PSPC-ACAL), Universidad de Buenos Aires, Argentina 2025
- **Poster Award**, Primer Congreso Chileno de Física de Plasmas 2023
- **Academic Excellence Scholarship**, Faculty of Sciences, Universidad de Chile 2022

TECHNICAL SKILLS

Scientific programming

C++ (advanced), Python (advanced), Bash (intermediate), Wolfram Mathematica (intermediate), MATLAB (basic). Simulation experience is in serial code; no prior MPI or GPU work.

Methods

Sandpile and cellular-automaton models; complex network analysis; magnetohydrodynamics; stochastic simulation; finite-size scaling; multifractal analysis; time-series and waiting-time statistics; power-law, exponential and extreme-value fitting with error estimation.

Solar and space physics data

SDO/HMI magnetograms; HEK flare catalogue; RHESSI hard X-ray data.

Languages

Spanish (native); English (fluent); Italian (basic); Portuguese (basic).

PROFESSIONAL MEMBERSHIPS

- Complex Systems Society (2025 – present)
- Asia-Pacific Plasma Physics Society (2025 – present)

REFEREES

- **Dr. Víctor Muñoz** — Professor, Department of Physics, Universidad de Chile. Doctoral advisor.
- **Dr. Denisse Pastén** — Department of Physics, Universidad de Chile. Doctoral advisor.
- **Dr. Fabio Lepreti** — Università della Calabria, Italy. Research-visit host.
- **Dr. Laura Morales** — INFIP, UBA-CONICET, Argentina. Co-author and research-visit host.